

Double-click (or enter) to edit

```
from google.colab import files
uploaded = files.upload()
```

KGE_5TKp...85GBjqb.jpg

KGE_5TKpQBe8dRmCoZqk3UCIKMvMukqQNJeAgtP-O4FkXm3-WYWdqUOXh4QnfIRBg36zdUIxzyWVANW4AxEIHmF3UbXRpu8SbZDC2prwOL0JePBtHSNNmsS2Oh9wnfhY04BUKzy1DXsQmAt2H6ul8YsN:zAUguV8eSx8U85GBjqb.jpg(image/jpeg) - 40963 bytes, last modified: 7/22/2026 - 100% done
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```
# Step 1: Install libraries
!pip install opencv-python matplotlib

# Step 2: Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Step 3: Upload image
from google.colab import files
uploaded = files.upload()

# Step 4: Read image
img_path = list(uploaded.keys())[0]
img = cv2.imread(img_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Step 5: Simulate Distortions
# -----

# (a) Add Gaussian Noise (Sensor noise)
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noise = np.random.normal(0, 25, img.shape).astype(np.uint8)
noisy_img = cv2.add(img, noise)

# (b) Blur (Optical defocus)
blur_img = cv2.GaussianBlur(noisy_img, (9,9), 0)

# (c) Downsample (Low resolution / sampling issue)
small = cv2.resize(blur_img, (img.shape[1]//4, img.shape[0]//4))
distorted_img = cv2.resize(small, (img.shape[1], img.shape[0]))

# -----
# Step 6: Improvements
# -----

# (a) Denoising
denoised = cv2.fastNlMeansDenoisingColored(distorted_img, None, 10, 10, 7, 21)

# (b) Sharpening (restore edges)
kernel = np.array([[0,-1,0],[-1,5,-1],[0,-1,0]])
sharpened = cv2.filter2D(denoised, -1, kernel)

# -----
# Step 7: Show Results
# -----

plt.figure(figsize=(12,6))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(distorted_img)
plt.title("Distorted Image")
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(sharpened)
```

```
plt.title("Improved Image")
plt.axis('off')

plt.show()
```

Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
 Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
 Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
 Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
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 Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.2)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib)

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mQtsrFlnIDjGwmdGSF3E5VTUtDhQfoQvLxiHVvpLHXIXWZWREtyMpxEdAmgW4ygP5q4sY57RFYmJqxqxIUl_lpumqUCkaA78b2s9wyGGhU
KA1inwq5XjlgRhGAFwZK9Y7AXLXnxFSvcZqgAO9Y.jpg(image/jpeg) - 18911 bytes, last modified: 7/22/2026 - 100% done

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Original Image



Distorted Image



Improved Image



